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CS-499: Computer Science Capstone

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Narratives

Briefly describe the artifact. What is it? When was it created?

Once again the selected artifact remains the same. It is the same event planner from the CS 360: Mobile Architecture and Programming class. This time, I focused on the database category enhancement, which replaced the local SQLite storage with a MySQL database. Instead of storing users and events only on the Android simulator device, the application now connects to a Node.js backend API, which then communicates with the MySQL database.

This is the most extensive and difficult enhancement, but it is the one that has made the app much more realistic. The API itself works as a middleman between the database and the front end.

The database itself has two tables, a user and an event one, both connected through a foreign key. Here is an image of the database titled “event_planner_db”:

```
[mysql> SHOW TABLES;
+-----+
| Tables_in_event_planner_db |
+-----+
| events                      |
| users                      |
+-----+
2 rows in set (0.00 sec)

mysql> 
```

As far as functionality goes, it still allows for full CRUD, which is something that was previously implemented but needed to be migrated.

That said, to run this version, one must use MAMP, which should be started so that the MySQL server is available. Then, the Node.js backend must also be started from the API folder using `npm start`. After that, one can run the application from Android Studio. The Android emulator will connect everything to the right port on the backend, and the backend will connect to the database. Overall, this enhancement highlights the use of a relational database, API communication, and user-specific persistence, which was not available with the local storage.

Detailed Instructions

Due to the complexity of this enhancement I'm choosing to leave some short, detailed instructions that will be added on the E-Portfolio as well as on a README during the final submission.

Requirements

1. **MAMP (current version)**
2. **Node.js (current version)**

3. MySQL through MAMP

4. Android Studio (latest version)

Database Setup

- Start MAMP and ensure that MySQL is running. The specific database for the project is titled “event_planner_db”.

Backend Setup

- Open a command line or PowerShell.
- Navigate to the project folder.
- Navigate to event planner api folder.
- From the folder run:
 - a. npm install
 - b. npm start
 - c. the backend will start running

Running the App

- Open the Android Studio app and select the right project folder.
- Start emulator.
- Everything will connect automatically.

Note: Ensure that before opening the app both MAMP and the backend are running.

Justify the inclusion of the artifact in your ePortfolio. Why did you select this item? What specific components of the artifact showcase your skills and abilities in software development? How was the artifact improved?

This artifact has been selected due to a lack of projects from the approved course list. It is one of the few that I still have fully available and functional. Furthermore, it was a good choice because all three enhancements could be implemented on it.

During this enhancement I completely overhauled and replaced the app's storage system. I changed the SQLite storage to a MySQL database connected through a Node.js and Express API. I took inspiration from a previous full-stack class in which we did the same process with a non-relational database. This new database still allows for the same functionality as before with the added bonus of persistence, which allows each user to have their own events separate from one another.

Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome-coverage plans?

I firmly believe that I've met the course outcome related to this module. This enhancement not only connects to the one focused on databases but it even touches on using computer science practices to solve problems. The newly connected database now demonstrates how data is managed in a more realistic and functional application.

Lastly, my outcome-coverage plan hasn't changed. The only slight deviation was the addition of a backend API which I didn't know was necessary until I did some research. Nevertheless, everything is starting to look good for the final submission.

Reflect on the process of enhancing and modifying the artifact. What did you learn as you were creating it and improving it? What challenges did you face?

I ran into a lot of issues during this enhancement. In fact, I originally wasn't even aware that I needed to create an API to use MySQL with the Android application. It might not be the most used or optimal solution, but I was already committed, which is why I still

implemented this solution. At first, I thought I could connect the app directly, but it actually needed a backend to work as a middleman with the database. Because of this, I had to dig deep through my GitHub and find my final project from the full stack class so I could relearn how to use Express and Node.js.

The database creation and testing was actually the easier part. MySQL itself is fairly simple, and there were many tutorials that helped. However, updating the XML files and changing the Java logic was a headache and required a lot of trial and error. In fact, the logic had to be rebuilt almost from scratch.

Nevertheless, I learned a great lesson, which is that database integration is much more complicated than just making a database and running queries on your app. If it weren't for my previous project from my full-stack class and experience from a bootcamp, I do not think I would have been able to do this enhancement.